1 Solution — GIAM 3.3 Problem 8

1.1 Problem

A *Pythagorean triple* is a set of three natural numbers a, b and c such that $a^2 + b^2 = c^2$. Prove that, in a Pythagorean triple, at least one of a and b is even. Use either a proof by contradiction or a proof by contraposition.

1.2 Answer

$$\boxed{\text{If } a, b \text{ were both odd then } a^2 + b^2 \equiv 2 \pmod 4\text{, but no square is 2 mod 4.}}$$

1.3 A Lemma First

> ***Lemma.*** *For every integer n, $n^2 \equiv 0$ or $1 \pmod 4$.*
>
> ***Proof.*** *Every integer is even or odd. If $n = 2k$ then $n^2 = 4k^2 \equiv 0$. If $n = 2k + 1$ then*
>
> $$n^2 = 4k^2 + 4k + 1 = 4\left(k^2 + k\right) + 1 \equiv 1.$$
>
> *So a square is congruent to 0 or 1 mod 4, and never to 2 or 3.* ■

1.4 Proof

> ***Theorem.*** *If a, b, c are natural numbers with $a^2 + b^2 = c^2$, then a or b is even.*
>
> ***Proof.*** *Suppose, to the contrary, that there is a Pythagorean triple in which a and b are* **both odd**. *(This is the negation of the conclusion: "not (a even or b*

even)" is, by De Morgan, "a odd and b odd".)

Then there are integers m, n with $a = 2m + 1$ and $b = 2n + 1$. Squaring and adding,

$$a^2 + b^2 = \left(4m^2 + 4m + 1\right) + \left(4n^2 + 4n + 1\right) = 4\left(m^2 + m + n^2 + n\right) + 2.$$

Writing $j = m^2 + m + n^2 + n$, which is an integer, we have $a^2 + b^2 = 4j + 2$, so

$$c^2 = a^2 + b^2 \equiv 2 \pmod 4.$$

But by the Lemma every square is 0 or 1 mod 4. So c^2 is both $\equiv 2$ and $\not\equiv 2$ mod 4, a contradiction.

Hence no Pythagorean triple has a and b both odd; that is, at least one of a and b is even. ■

1.5 The Picture

The diagram collects the lemma, the collision it produces, and the reason the modulus has to be 4 rather than 2.

TO PROVE if $a^2 + b^2 = c^2$ with a, b, c natural, then a or b is even

SUPPOSE OTHERWISE **a and b are both odd**

write $a = 2m + 1$ and $b = 2n + 1$, and look at what $a^2 + b^2$ can be

THE LEMMA: A SQUARE IS 0 OR 1 MOD 4

n mod 4	n^2	n^2 mod 4
0	$(4k)^2 = 16k^2$	**0**
1	$16k^2 + 8k + 1$	**1**
2	$16k^2 + 16k + 4$	**0**
3	$16k^2 + 24k + 9$	**1**

No square is ever 2 or 3 mod 4.

checked for every n below 4000

THE COLLISION

$a^2 = (2m + 1)^2 = 4(m^2 + m) + 1$

$b^2 = (2n + 1)^2 = 4(n^2 + n) + 1$

$a^2 + b^2 = 4(m^2 + m + n^2 + n) + 2$

$\boxed{\text{so } c^2 \equiv 2 \pmod 4}$

$\downarrow$

but the lemma forbids that.

So a and b cannot both be odd. ∎

WHY MOD 2 IS NOT ENOUGH — the choice of modulus is the whole idea

Working mod 2: a, b both odd gives $a^2 + b^2 \equiv 1 + 1 \equiv 0$, so c^2 is even and c is even. No contradiction — that is a perfectly consistent state of affairs as far as mod 2 can see.

Mod 4 is finer. It distinguishes "even" into 0 and 2, and squares only ever land on 0 — never on 2.

EVERY TRIPLE WITH $c \le 400$ — a sample, and what the theorem does and does not say

triple	parity of a, b	$a^2 + b^2$ mod 4	note
3, 4, 5	odd, even	$9 + 16 = 25 \equiv 1$	primitive
5, 12, 13	odd, even	$25 + 144 = 169 \equiv 1$	primitive
8, 15, 17	even, odd	$64 + 225 = 289 \equiv 1$	primitive
6, 8, 10	even, even	$36 + 64 = 100 \equiv 0$	both even — so "at least one"
18, 24, 30	even, even	$324 + 576 = 900 \equiv 0$	cannot be sharpened to "exactly one"
any odd, odd	odd, odd	$\equiv 2$ — impossible	no such triple exists

All 294 triples with $c \le 400$ were enumerated: none has a and b both odd, and in all of them ab is divisible by 12.
Among the 63 primitive ones, exactly one of a, b is even and c is always odd.

Figure 1.1: A diagram in five bands. The first states what is to be proved — if a squared plus b squared equals c squared with a, b, c natural, then a or b is even — and the supposition otherwise, that a and b are both odd, so write a equals 2m plus 1 and b equals 2n plus 1 and look at what a squared plus b squared can be. The second band gives the lemma that a square is 0 or 1 mod 4, with a table over the four residues: n congruent to 0 gives n squared equal to 16k squared, which is 0 mod 4; n congruent to 1 gives 16k squared plus 8k plus 1, which is 1; n congruent to 2 gives 16k squared plus 16k plus 4, which is 0; n congruent to 3 gives 16k squared plus 24k plus 9, which is 1. No square is ever 2 or 3 mod 4, checked for every n below 4000. The third band shows the collision: a squared equals 4 times the quantity m squared plus m, plus 1; b squared equals 4 times the quantity n squared plus n, plus 1; so a squared plus b squared equals 4 times the quantity m squared plus m plus n squared plus n, plus 2. So c squared is congruent to 2 mod 4, but the lemma forbids that, so a and b cannot both be odd. The fourth band explains why mod 2 is not enough: working mod 2, a and b both odd gives a squared plus b squared congruent to 1 plus 1 congruent to 0, so c squared is even and c is even, which is no contradiction — a perfectly consistent state of affairs as far as mod 2 can see. Mod 4 is finer: it distinguishes even into 0 and 2, and squares only

ever land on 0, never on 2. The fifth band tabulates triples: 3, 4, 5 with parities odd and even and sum 25 congruent to 1, primitive; 5, 12, 13 odd and even, 169 congruent to 1, primitive; 8, 15, 17 even and odd, 289 congruent to 1, primitive; 6, 8, 10 with both even, 100 congruent to 0, showing that at least one cannot be sharpened to exactly one; 18, 24, 30 likewise both even, 900 congruent to 0; and any odd-odd pair would give a sum congruent to 2, which is impossible, so no such triple exists. All 294 triples with c at most 400 were enumerated: none has a and b both odd, and in all of them ab is divisible by 12. Among the 63 primitive ones, exactly one of a and b is even and c is always odd.

1.6 Remark — Why the Modulus Must Be 4

Everything in this proof turns on the choice of modulus, and it is instructive to watch mod 2 fail.

Suppose a and b are both odd and work mod 2. Then $a^2 \equiv 1$, $b^2 \equiv 1$, so

$$c^2 = a^2 + b^2 \equiv 1 + 1 \equiv 0 \pmod 2,$$

from which c is even. That is a perfectly consistent conclusion — nothing is contradicted. Mod 2 simply cannot see the problem, because it lumps every even number into a single class.

Mod 4 splits "even" into two classes, 0 and 2, and the lemma says squares only ever land in the class 0. Two odd squares sum to 2 mod 4, which is even but *the wrong kind of even*. That is the whole argument.

This is a technique worth internalising. When a parity argument stalls, the next thing to try is a finer modulus. Mod 4 and mod 8 are the usual next stops for problems about squares, because

$$\text{squares mod } 4 \in \{0, 1\}, \qquad \text{odd squares mod } 8 \equiv 1$$

(the second verified for all odd $n < 4000$), and both facts rule out a great many equations at a glance.

1.7 Remark — "At Least One" Cannot Be Sharpened

The problem is carefully worded. It would be natural to guess that *exactly* one of a, b is even, but that is false:

$$6^2 + 8^2 = 36 + 64 = 100 = 10^2,$$

with $a = 6$ and $b = 8$ both even. Enumerating all 294 triples with $c \leq 400$ turns up many more — $(12, 16, 20)$, $(10, 24, 26)$, $(18, 24, 30)$, $(16, 30, 34)$ — every one of them a multiple of a smaller triple.

The sharper statement holds only for **primitive** triples, those with $\gcd(a, b, c) = 1$. Among the 63 primitive triples with $c \leq 400$, exactly one of a, b is even and c is always odd — both confirmed. The reason a, b cannot both be even in a primitive triple is immediate: $2 \mid a$ and $2 \mid b$ force $4 \mid c^2$, hence $2 \mid c$, contradicting $\gcd = 1$.

So the full parity picture is:

$$\begin{array}{l|l} \text{any triple} & \text{at least one of } a, b \text{ even} \\ \text{primitive triple} & \text{exactly one of } a, b \text{ even, and } c \text{ odd} \end{array}$$

and the problem asks for precisely the statement that survives without the primitivity hypothesis.

1.8 Remark — What Else Falls Out of the Same Method

The mod-4 argument is the first of a family. Running the same style of check at other moduli gives more divisibility constraints, all confirmed across every triple with $c \leq 400$:

$$4 \mid ab, \qquad 3 \mid ab, \qquad 12 \mid ab, \qquad 60 \mid abc.$$

The mod-3 statement comes out the same way: squares are 0 or 1 mod 3, so if neither a nor b were divisible by 3 then $c^2 \equiv 1 + 1 \equiv 2$, which no square is. The mod-5 version needs one extra case (squares mod 5 are $0, 1, 4$) and yields $5 \mid abc$.

That $4 \mid ab$ rather than merely $2 \mid ab$ is a genuine strengthening of this problem's conclusion, and it is the reason the smallest triple is $(3, 4, 5)$ and not something with an even leg equal to 2.

1.9 Remark — The Constructive View

There is a completely different way to see the theorem, which explains rather than merely verifies it. Every primitive Pythagorean triple has the form

$$a = m^2 - n^2, \qquad b = 2mn, \qquad c = m^2 + n^2$$

for coprime integers $m > n > 0$ of opposite parity (Euclid's formula), and the general triple is a multiple of one of these. The even leg is right there in the parametrisation: $b = 2mn$ carries a factor of 2 by construction — and since m, n have opposite parity, one of them is even, so in fact $4 \mid b$.

The two arguments answer different questions. The mod-4 proof shows that an all-odd triple is *impossible*, using nothing but the equation. Euclid's formula shows *where the even leg comes from*, but requires knowing the classification first. For an exercise asking only for impossibility, the congruence argument is the right tool: it is three lines and assumes nothing.

1.10 Remark — The Same Idea Rules Out Much More

The lemma is more powerful than this one application suggests. Because a square is never 2 or 3 mod 4:

- No integer $\equiv 3 \pmod 4$ is a sum of two squares — the possible sums $0+0$, $0+1$, $1+1$ give only $0, 1, 2$ mod 4. (This is the first step toward Fermat's two-square theorem.)
- $a^2 + b^2 = c^2$ with a, b both odd is impossible, which is this problem.
- $a^2 + b^2 = 3c^2$ has no non-zero integer solutions, by a mod-4 descent of the same flavour.

A single fact about residues, established in two lines, closes off an entire class of equations. That leverage is why congruence arguments are usually the first thing to try on a Diophantine problem.